NutriCart Hub - Mega Summary

1. What problem are we actually addressing?

Original framing:

> *How might context-aware mobile technology and social features support UQ students in making healthier food choices while grocery shopping?*

What the research pushed us toward:

> *How might we support students' real grocery decisions around price, time, personal priorities and household needs without adding effort, judgement or another unnecessary app?*

Important point to remember:

- We are not trying to define one universal version of healthy shopping.
- The project is about supporting decision-making, not rating or policing food choices.
- A good outcome is not that every original feature survives. A good outcome is that the design changes can be traced back to research.

4. Research and prototype timeline

1. Exploratory interviews: Learned about price, time, convenience, familiarity, nutrition priorities and social influence.

2. Social Shopping Companion App: Used as a low-fidelity design probe with friend recommendations, group goals, recipes and a shopping list.

3. Prototype interviews: Found useful parts, but also problems with missing price information, social comparison, logging effort, timing and the idea of another standalone app.

4. Thematic synthesis: Grouped recurring findings and reframed the problem around practical grocery decisions instead of a single idea of healthy eating.

5. Divergent ideation: Explored physical concepts such as a budget device, shopping traffic-light idea, shared list device and trolley-based support.

6. NutriCart Hub: Combined the strongest ideas into a Home Dock plus detachable Cart Unit.

7. Latest prototype testing: Tested digital/HTML and physical versions. Budget, shared lists and trusted recommendations remained strong; size, data entry, speed and alerts need work.

8. Attempted shadowing: Low participation exposed privacy, self-consciousness, time burden and observer-reactivity issues.

9. Next method: Move toward lower-burden field use with a short retrospective interview rather than following participants around the store.

5. Main themes from the interviews

Price and budget structure the shop: Students repeatedly talked about weekly budgets, specials, cheaper brands, price per 100 g and wanting price information in the prototype.

Time and convenience can override the plan: When rushed, people described choosing whatever was nearby, quicker to prepare or easier to find, even when it was not what they originally planned.

"Healthy" is subjective: Participants used different criteria such as protein, sugar, calories, fat, sodium, ingredients, fresh produce, visible vegetables or simply whether something seemed less greasy. Some did not care much about health at all.

Familiarity and routine reduce effort: People often buy the same brands and foods repeatedly. They are more likely to compare products when something is unfamiliar or when price changes.

Social influence depends on trust and relevance: Friends, partners and housemates can change what someone buys, but recommendations are more useful when they come from a person with similar needs, budget or habits.

Social comparison is divisive: Some users find competition motivating. Others feel pressure, judgement or financial unfairness. This is why comparison should not be a default part of the experience.

Coordination is more consistently useful than a social feed: Shared lists, housemate requests, knowing what is already at home and coordinating who buys what were repeatedly useful.

Support has to appear at the right moment: Planning information is more useful before shopping. In-store information needs to be short and actionable.

Another app can become part of the problem: Some participants preferred retailer apps or disliked having to log food and manage another separate app.

6. What NutriCart Hub is now

NutriCart Hub is a shared household grocery-planning device with two parts.

Home Dock

- Lives in the kitchen or another shared household space.
- Shared shopping list.
- Meal ideas with practical details such as cost, time and servings.
- Housemate requests.
- What is already at home.
- Rough shopping budget.
- Optional social features such as trusted recommendations or shared goals.

Cart Unit

- Detaches from the Home Dock and goes to the supermarket.
- Shopping checklist as the main screen.
- Running budget / amount remaining.
- Optional product comparison.
- Rare deal and already-at-home alerts.
- Can add off-list items directly while shopping.

Core principle:

> *The system provides useful information, but the shopper decides what matters. It does not tell them which product is universally "healthy" or "better".*

7. What the latest prototype round told us

Things that are worth keeping:

- Shared shopping list.
- Running budget.
- Trusted friend/housemate recommendations.
- Neutral product comparison without a health verdict.
- Simple, minimal in-store interaction.

Things that need to change:

- Make the Cart Unit smaller and easier to use with both trolleys and baskets.
- Make the shopping list the default Cart Unit view.
- Make the remaining budget very visible.
- Keep comparison optional, not part of every purchase.
- Do not make users scan and compare if it creates noticeable delay.
- Reduce alert frequency. Prioritise deals and already-at-home reminders.
- Make Home Dock input faster and easier.
- Allow specific brands when they matter, but do not require brand detail for every item.
- Allow spontaneous/off-list purchases directly from the Cart Unit.
- Keep group goals optional because users respond to them differently.

8. Easy "finding -> design change" talking points

Research finding	Design response
Users care about price	Budget and price information became core features.
Users shop quickly	The Cart Unit is simplified and comparison is optional.
Healthy means different things	Comparison shows information instead of giving a healthy/unhealthy verdict.
Friend recommendations can help	Recommendations are kept, but focused on trusted people rather than a public feed.
Social comparison can create pressure	Leaderboard-style health comparison was removed from the main flow.
People already use lists	The shared list became one of the main functions.
Housemates/partners affect purchases	The Home Dock supports shared requests and household coordination.
People forget what is already at home	Added already-at-home reminders.
People make spontaneous purchases	Add Item is available on the Cart Unit.
Manual input feels like work	Next prototype should test quick add, barcode scan or simulated voice input.
Another app was not wanted by everyone	The design moved toward a dedicated physical household/in-store concept.

9. HCI concepts we can refer to

User-centred design: The design changes in response to what users actually need and reject.

Low-fidelity prototyping: Cardboard, paper and HTML prototypes let us test ideas before implementing electronics.

Design probe: The original app was useful because it gave people something concrete to react to, even though it was not the final product.

Think-aloud: Participants described what they noticed, understood and disliked while using the prototype.

Thematic analysis / affinity-style synthesis: We grouped repeated codes into broader patterns such as budget, time pressure, trust, coordination and interaction effort.

Iterative design: Negative feedback led to removal or reframing of features instead of just visual polishing.

Context of use: The Home Dock and Cart Unit separate planning at home from fast decision-making in-store.

Ecological validity: We tried to understand the prototype in a realistic shopping context rather than only asking hypothetical questions.

Reactivity: Being watched during shopping may change behaviour, which became a limitation of shadowing.

Participant burden: Shadowing required more time and felt more intrusive than an interview for many potential participants.

Triangulation: The next stage should combine what users said, what they actually used, and how they explain that use afterwards.

10. Research limitations we should be honest about

- Some early prototype questions were leading or explained the intended benefit before the participant fully reacted.
- Some interviews were detailed transcripts while others were group notes, so we should be cautious with exact counts and percentages.
- Not every participant saw the exact same prototype version in the latest round.
- Because HTML and physical versions were not standardised conditions, we should not claim one version caused better responses than the other.
- People who agreed to be interviewed may already be more willing to discuss food and shopping than people who declined.
- Shadowing created privacy, self-consciousness and time concerns, and the observer may change how the participant shops.

Safe wording if challenged:

> *We are treating these sessions as formative qualitative research rather than a controlled comparison. The goal is to identify recurring needs, problems and design questions, then test them again in later iterations.*

11. Why shadowing did not work well, and what we do next

The low participation in shadowing is useful evidence about the method. Grocery shopping involves personal food choices, budgets, habits and household information. Some people did not want an observer present, and others did not have time for a longer session. The researcher may also change the behaviour being observed.

Next method:

1. Give each participant the same prototype and a short neutral instruction.
2. Let them use it during a normal shopping trip without a researcher following them.
3. Record only simple prototype interactions where possible, such as opening the list, checking budget, using Compare, dismissing an alert or adding an item.
4. Do a short post-shop retrospective interview using the prototype state or receipt as a memory prompt.

Neutral prompts we can use:

- What did you use first?
- Was there anything you ignored?
- What made you use Compare there?
- Was there anything that slowed you down?
- Was there anything you wanted the prototype to do but it could not?
- What would you remove?
- What would you normally do instead of using this?

12. Likely questions we might be asked, with safe answers

Q: Why did you move away from the original app?

A: The app worked as a design probe, but the interviews showed that social interaction was not the only or strongest need. Price, time, existing lists, household coordination and another-app burden came up repeatedly, so we moved toward a physical concept that fits those needs better.

Q: Why is this not just a shopping list?

A: The shared list is the core because it matches existing behaviour, but the design also links home planning to in-store support through budget tracking, household requests, optional comparison and contextual reminders.

Q: Why have a physical device instead of a phone?

A: This is still a design question we are testing. One reason for exploring a physical device was that some users did not want another standalone app. The physical prototype also lets us test whether a dedicated shared object fits better into household planning and the trolley context.

Q: What does social computing mean in your project now?

A: The social part has shifted from public comparison to practical coordination between trusted people: shared lists, household requests, meal ideas and relevant recommendations.

Q: Are you trying to make people eat healthier?

A: Not in the sense of imposing one definition of healthy. Our research showed that people use very different criteria. The current design provides information and supports the user's own priorities.

Q: Why keep group goals if they caused problems?

A: We do not treat them as a core feature. Some participants found them motivating while others found them stressful. The next iteration would make them optional and test different framings, such as household or budget goals instead of only health goals.

Q: What is the strongest feature right now?

A: The most consistently practical functions are the shared list and budget support. Trusted recommendations are also useful when they are relevant, but they should not become a constant feed.

Q: What is the biggest unresolved problem?

A: Whether people will maintain the Home Dock with low enough effort, and whether the Cart Unit adds enough value without slowing a normal shop.

Q: Why did shadowing fail?

A: It was not only recruitment. The method itself was intrusive for some people. That tells us to use a lower-burden field method that still gives us behavioural evidence.

Q: What will you test next?

A: Whether the simplified Cart Unit is actually used during a normal shop, which functions are ignored, whether comparison is fast enough, whether alerts are useful rather than annoying, and how little effort is needed to keep household information updated.

13. Things we should not overclaim

- Do not say every participant wanted the same thing.
- Do not say group goals were proven to work.
- Do not say the physical prototype was proven better than the HTML version.
- Do not say the system will make people healthier.
- Do not say shadowing failed only because we had bad luck recruiting.
- Do not say product comparison should happen for every item.
- Do not present a participant suggestion as a final requirement unless it also makes sense with the broader findings.
- Avoid exact percentages unless we have checked the participant count and prototype exposure carefully.

14. Team alignment: the points everyone should agree on

- The project is no longer mainly about a social healthy-eating app.
- The current problem is practical grocery decision support under real constraints such as price, time and household needs.
- The original app was still useful because it exposed assumptions and gave users something concrete to react to.
- The social component is now about trusted coordination, not public performance.
- The current prototype should become simpler, not more feature-heavy.

- The Home Dock is for planning; the Cart Unit is for quick in-store use.
- The shared list and budget are the strongest core functions at the moment.
- Comparison is optional and should remain neutral.
- Alerts should be rare and useful.
- The next research method should reduce privacy concerns and participant burden.
- We should be open about limitations and avoid overclaiming from small qualitative samples.

15. Emergency presentation script

If someone has to explain the whole project with almost no preparation, use this order:

1. Start with the problem: We began by looking at how students make grocery choices and whether social/context-aware technology could support healthier decisions.

2. Explain what the first interviews changed: We found that price, time, convenience, routines and different definitions of healthy were just as important, if not more important, than social motivation.

3. Explain the first prototype: We made a Social Shopping Companion App with friend recommendations, group goals, recipes and a shopping list as a design probe.

4. Explain the negative feedback: Some users liked friend recipes, but price was missing, logging felt like effort, social comparison could feel judgemental, and some people did not want another app.

5. Explain the reframing: So we stopped asking how to make a better healthy-shopping app and instead asked how to support real grocery decisions without judgement or extra effort.

6. Explain NutriCart Hub: The Home Dock handles planning and household coordination. The Cart Unit handles the list, budget, optional comparison and rare alerts in-store.

7. Explain the latest feedback: The newest testing supports the list, budget and trusted recommendations, but tells us to make the device smaller, the interaction faster, alerts rarer and data entry easier.

8. Explain the method limitation: We tried to move toward shadowing, but many people found grocery observation intrusive or too time-consuming, so the method itself became part of the research lesson.

9. Finish with the next step: Next we want lower-burden field testing where users shop normally with the simplified prototype and then explain what they actually used or ignored afterwards.

16. One-line mental map

> *Interviews -> app probe -> problems exposed -> thematic synthesis -> problem reframed -> physical Hub -> usability testing -> simplify -> field-method rethink -> next iteration.*

Final reminder: If a team member forgets a detail, come back to the evidence. Explain what users said or did, what design assumption that challenged, and what changed because of it.